Gebze Technical University

Department of Mechanical Engineering

Recitation 2

Equivalent Elements and Harmonic Motion

Lecturer: Saeed Lotfan | saeedlotfan.com

Self-study guide. These 32 questions develop the Week 2 topics through explanations and worked calculations. Each question is followed immediately by its solution.

Levels: Foundation builds the basic relations; Application uses them in a specified model; Challenge combines several steps or requires closer interpretation. The questions progress within each topic. Angles are in radians unless degrees are stated.

Question 1 | Springs in parallel and in series — Foundation

Find the equivalent stiffness of two linear springs of stiffnesses k_1 and k_2 connected (a) in parallel and (b) in series. How should several springs be connected to increase the overall stiffness?

Solution 1

1. State what equivalence means

An equivalent spring has the same terminal force F for the same terminal displacement x: $F = k_{\text{eq}}x$. Assume linear springs and negligible mass at connecting points.

2. Parallel connection

Both springs undergo the same extension, $x_1 = x_2 = x$, and their forces add:

$$F = F_1 + F_2 = k_1x + k_2x = (k_1 + k_2)x.$$

Therefore $\boxed{k_{\text{eq}} = k_1 + k_2}$. More parallel springs increase stiffness, provided their attachment points undergo the same displacement.

3. Series connection

The massless intermediate connection transmits the same force through both springs. Their extensions are $x_1 = F/k_1$ and $x_2 = F/k_2$. The total extension is

$$x = x_1 + x_2 = F\left(\frac{1}{k_1} + \frac{1}{k_2}\right).$$

Thus

$$\boxed{\frac{1}{k_{\text{eq}}} = \frac{1}{k_1} + \frac{1}{k_2}}, \qquad \boxed{k_{\text{eq}} = \frac{k_1k_2}{k_1 + k_2}}.$$

For positive stiffnesses, the series combination is softer than either component. Every stiffness in these equations has units N/m.

Question 2 | Four-spring combinations — Application

Four springs have stiffnesses

$$k_1 = 3500, \quad k_2 = 8500, \quad k_3 = 17500, \quad k_4 = 35000 \quad \text{N/m}.$$

Find the equivalent stiffness for each connection:

(a) All four in parallel.

(b) All four in series.

(c) k_1, k_2 in parallel; denote the result k_{12}.

(d) k_3, k_4 in parallel; denote the result k_{34}.

(e) k_1, k_2, k_3 in parallel; denote the result k_{123}.

(f) k_{123} in series with k_4.

(g) k_2, k_3, k_4 in parallel; denote the result k_{234}.

(h) k_1 in series with k_{234}.

Solution 2

Use a sum for a parallel connection and a sum of reciprocals for a series connection. Reduce each stated group before combining it with another group.

(a) $k_{\text{eq}} = 3500 + 8500 + 17500 + 35000 = \boxed{64500 \text{ N/m}}$.

(b) $k_{\text{eq}} = (1/3500 + 1/8500 + 1/17500 + 1/35000)^{-1} = \boxed{2044.674 \text{ N/m}}$.

(c) $k_{12} = 3500 + 8500 = \boxed{12000 \text{ N/m}}$.

(d) $k_{34} = 17500 + 35000 = \boxed{52500 \text{ N/m}}$.

(e) $k_{123} = 3500 + 8500 + 17500 = \boxed{29500 \text{ N/m}}$.

(f) $k_{\text{eq}} = \dfrac{29500(35000)}{29500 + 35000} = \boxed{16007.752 \text{ N/m}}$.

(g) $k_{234} = 8500 + 17500 + 35000 = \boxed{61000 \text{ N/m}}$.

(h) $k_{\text{eq}} = \dfrac{3500(61000)}{3500 + 61000} = \boxed{3310.078 \text{ N/m}}$.

The answers in (b), (f), and (h) are each smaller than the smallest stiffness connected in that series stage. This is a useful check on the reduction.

Question 3 | Freight-truck suspension **Application**

The suspension of a freight truck uses three identical helical springs in a parallel arrangement. Each spring is made of steel with shear modulus $G = 80 \times 10^9\,\mathrm{N/m^2}$, has $n = 5$ effective turns, mean coil diameter $D = 0.20\,\mathrm{m}$, and wire diameter $d = 0.02\,\mathrm{m}$. Find the equivalent spring constant of the suspension.

Solution 3

1. Find the stiffness of one spring

For a close-coiled helical spring in axial loading, with n effective turns,

$$k = \frac{Gd^4}{8D^3n}.$$

This model neglects direct shear and assumes elastic deformation. Here

$$d^4 = (0.02)^4 = 1.60 \times 10^{-7}\ \mathrm{m}^4, \qquad D^3 = (0.20)^3 = 8.00 \times 10^{-3}\ \mathrm{m}^3.$$

Substitution gives

$$k = \frac{(80 \times 10^9)(1.60 \times 10^{-7})}{8(8.00 \times 10^{-3})(5)} = 40000\ \mathrm{N/m}.$$

The units are $(\mathrm{N/m^2})(\mathrm{m}^4)/\mathrm{m}^3 = \mathrm{N/m}$.

2. Combine the three springs

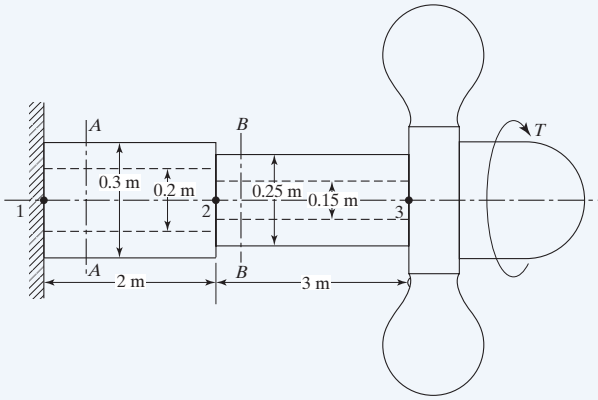
The parallel arrangement gives all three springs the same compression x. Hence

$$F = 3kx = k_{\mathrm{eq}}x, \qquad \boxed{k_{\mathrm{eq}} = 3(40000) = 120000\ \mathrm{N/m}}.$$

This sum assumes the supporting members move together without rotation that would give the springs different compressions.

Question 4 | Two-segment propeller shaft — Application

Determine the torsional spring constant of the steel propeller shaft shown below. The shaft has two uniform segments in series: segment 12 with length $l_{12} = 2\,\text{m}$, outer/inner diameters $D_{12} = 0.30\,\text{m}$ and $d_{12} = 0.20\,\text{m}$, and segment 23 with length $l_{23} = 3\,\text{m}$, diameters $D_{23} = 0.25\,\text{m}$ and $d_{23} = 0.15\,\text{m}$. Take the shear modulus $G = 80 \times 10^9\,\text{N/m}^2$.



Solution 4

1. Calculate the polar second moments of area

For a hollow circular section, $J_p = \pi(D^4 - d^4)/32$. Therefore

$$J_{12} = \frac{\pi}{32}(0.30^4 - 0.20^4) = 6.38136 \times 10^{-4}\ \text{m}^4,$$

$$J_{23} = \frac{\pi}{32}(0.25^4 - 0.15^4) = 3.33794 \times 10^{-4}\ \text{m}^4.$$

These are area moments, not mass moments of inertia.

2. Calculate each torsional stiffness

For a uniform elastic shaft segment, $\theta_i = Tl_i/(GJ_i)$, so $k_{ti} = GJ_i/l_i$:

$$k_{t12} = \frac{80 \times 10^9(6.38136 \times 10^{-4})}{2} = 25.52544 \times 10^6\ \text{N m/rad},$$

$$k_{t23} = \frac{80 \times 10^9(3.33794 \times 10^{-4})}{3} = 8.90118 \times 10^6\ \text{N m/rad}.$$

3. Add the twists under the common torque

The same torque passes through both segments; the end twist is

$$\theta = \theta_{12} + \theta_{23} = T\left(\frac{1}{k_{t12}} + \frac{1}{k_{t23}}\right).$$

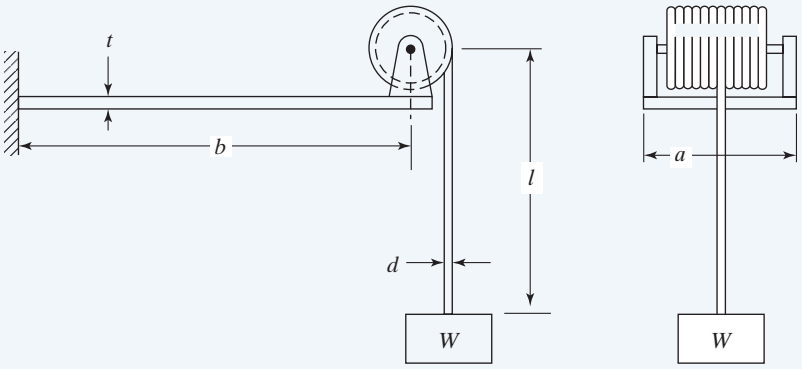
Thus

$$\boxed{k_{t,\text{eq}} = \left(\frac{1}{k_{t12}} + \frac{1}{k_{t23}}\right)^{-1} = 6.59973 \times 10^6\ \text{N m/rad}}.$$

The result is smaller than either segment stiffness, as expected for torsional springs in series.

Question 5 | Hoist supported by a cantilever — Challenge

A hoisting drum carrying a steel wire rope is mounted at the end of a cantilever beam. Let the beam length be b, its rectangular cross–section be $a \times t$, and the suspended rope length be l. The rope has net diameter d. Assume the Young's modulus of both the beam and the rope is E. Determine the equivalent spring constant seen at the load W.



Solution 5

1. Identify both contributions to load displacement

The load moves downward because the beam tip bends and the rope stretches. Neglect the drum's compliance and any unwinding. Use small elastic deflections and a massless-support model.

2. Obtain the beam and rope stiffnesses

The beam bends about the axis for which $I = at^3/12$. Its tip stiffness is

$$k_b = \frac{3EI}{b^3} = \frac{Eat^3}{4b^3}.$$

The effective rope area is $A_r = \pi d^2/4$, so its axial stiffness is

$$k_r = \frac{EA_r}{l} = \frac{E\pi d^2}{4l}.$$

3. Apply the common force and add deflections

An increment of load force W causes

$$\delta_b = \frac{W}{k_b}, \qquad \delta_r = \frac{W}{k_r}.$$

The load-point deflection is $x = \delta_b + \delta_r$. Therefore these compliances act in series:

$$\frac{1}{k_{\text{eq}}} = \frac{x}{W} = \frac{4b^3}{Eat^3} + \frac{4l}{E\pi d^2}.$$

Combining the fractions gives

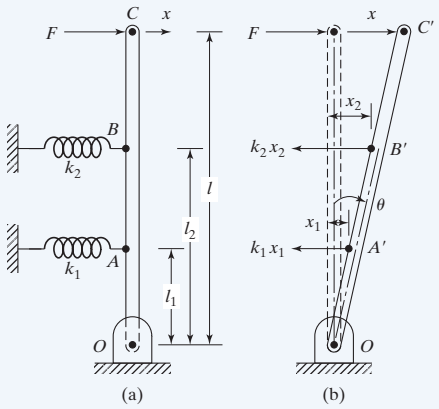
$$\boxed{k_{\text{eq}} = \frac{E\pi at^3 d^2}{4(\pi d^2 b^3 + lat^3)}}.$$

4. Check the result

If the beam becomes rigid, $1/k_b \to 0$ and $k_{\text{eq}} \to k_r$. If the rope becomes rigid, $k_{\text{eq}} \to k_b$. Both limits agree with the physical model.

Question 6 | Stiffness referred to a bar tip — Application

A rigid bar of length l is hinged at O and connected to two linear springs of stiffnesses k_1 and k_2 attached at distances l_1 and l_2 from the hinge. A horizontal force F is applied at the free end C producing a small horizontal displacement x at C. Find the equivalent spring constant k_{eq} such that $F = k_{\text{eq}}\, x$.



Solution 6

1. Relate every spring extension to the chosen coordinate

For a small bar rotation θ, the horizontal displacements satisfy

$$x = l\theta, \qquad x_1 = l_1\theta = \frac{l_1}{l}x, \qquad x_2 = l_2\theta = \frac{l_2}{l}x.$$

The spring extensions differ unless $l_1 = l_2$.

2. Match the elastic energy

The two springs store

$$U = \tfrac{1}{2}k_1x_1^2 + \tfrac{1}{2}k_2x_2^2 = \tfrac{1}{2}\left[k_1\left(\frac{l_1}{l}\right)^2 + k_2\left(\frac{l_2}{l}\right)^2\right]x^2.$$

Equating this to $\frac{1}{2}k_{\text{eq}}x^2$ gives

$$\boxed{k_{\text{eq}} = k_1(l_1/l)^2 + k_2(l_2/l)^2}.$$

3. Verify by moments

Moment balance about O gives $Fl = k_1x_1l_1 + k_2x_2l_2$. Substitute the displacement relations and divide by l to recover $F = k_{\text{eq}}x$.

4. Interpret the coordinate dependence

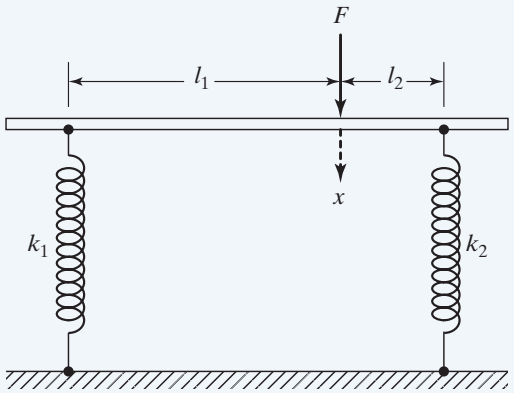
If the coordinate is θ, the equivalent rotational stiffness is

$$K_\theta = k_1l_1^2 + k_2l_2^2, \qquad k_{\text{eq}} = K_\theta/l^2.$$

K_θ has units N m/rad and k_{eq} has units N/m. Both describe the same physical springs at different reference coordinates.

Question 7 | A supported bar that can rotate — Challenge

Two vertical springs with stiffnesses k_1 (left) and k_2 (right) support a rigid, massless bar. A vertical force F is applied at a point between the springs such that the distances from the load point to the left and right spring attachments are l_1 and l_2, respectively. The bar is horizontal when $F = 0$, but under load it may rotate. Let x be the vertical displacement of the load point. Determine the equivalent spring constant k_{eq} defined by $F = k_{\text{eq}}\, x$.



Solution 7

1. Find the spring forces by statics

Let $L = l_1 + l_2$ and let R_1, R_2 be the upward forces on the bar. Force balance and moments about the left attachment give

$$R_1 + R_2 = F, \qquad R_2 L = F l_1.$$

Thus $R_1 = F l_2/L$ and $R_2 = F l_1/L$.

2. Find the endpoint displacements

Taking downward displacements as positive,

$$y_1 = \frac{R_1}{k_1} = \frac{F l_2}{k_1 L}, \qquad y_2 = \frac{R_2}{k_2} = \frac{F l_1}{k_2 L}.$$

There is no constraint requiring $y_1 = y_2$.

3. Interpolate the displacement along the rigid bar

For small rotation, displacement varies linearly along the bar. The force acts a distance l_1 from the left attachment, so

$$x = y_1 + \frac{l_1}{L}(y_2 - y_1) = \frac{l_2}{L} y_1 + \frac{l_1}{L} y_2.$$

Substitution yields

$$x = \frac{F}{L^2}\left(\frac{l_2^2}{k_1} + \frac{l_1^2}{k_2}\right).$$

Therefore

$$\boxed{k_{\text{eq}} = \frac{(l_1 + l_2)^2}{l_2^2/k_1 + l_1^2/k_2} = \frac{k_1 k_2 (l_1 + l_2)^2}{k_2 l_2^2 + k_1 l_1^2}}.$$

4. Explain why a simple parallel sum can fail

The springs look parallel, but the bar can rotate. They have equal displacement only when $k_2 l_2 = k_1 l_1$. In that special case $F = (k_1 + k_2)x$; otherwise the coordinate compatibility above is essential.

Question 8 | Equivalent inertia and coordinates — Foundation

State whether each statement is true or false and explain your answer.

(a) The equivalent mass of several masses at different locations can be found using equivalence of kinetic energy.

(b) Generalized coordinates are not necessarily Cartesian coordinates.

Solution 8

1. Equivalent mass: true, with the coordinate specified

For a one-coordinate system, express every body velocity in terms of $\dot{q}$. If $v_i = a_i\dot{q}$ and $\omega_j = b_j\dot{q}$, with constant kinematic ratios,

$$T = \tfrac{1}{2}\sum_i m_i a_i^2\dot{q}^2 + \tfrac{1}{2}\sum_j J_j b_j^2\dot{q}^2.$$

Equating this to $\frac{1}{2}M_{\text{eq}}\dot{q}^2$ gives

$$M_{\text{eq}} = \sum_i m_i a_i^2 + \sum_j J_j b_j^2.$$

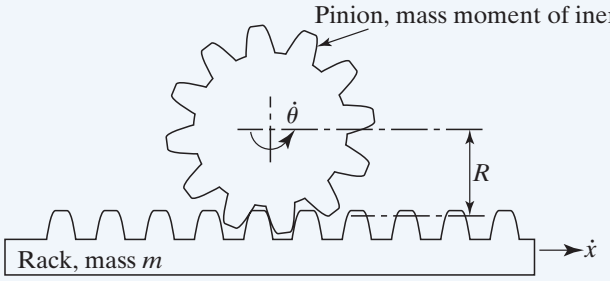
For a translational q, this coefficient is an equivalent mass. For an angular q, it is an equivalent rotational inertia. Bodies moving at different speeds generally cannot be replaced by their physical mass sum. If more than one coordinate is independent, a single scalar equivalent mass is generally insufficient.

2. Generalized coordinates: true

A coordinate is any independent variable that specifies configuration after constraints are imposed. A pendulum can use its angle θ instead of Cartesian coordinates. With a fixed length l, its bob position is determined by $x = l\sin\theta$ and $y = -l\cos\theta$ (upward y positive). Thus one angle describes the motion.

Question 9 | Rack-and-pinion equivalent inertia — Application

A rack of mass m translates without slip against a pinion of radius R and mass moment of inertia J_0 about its fixed axis. Find (a) the equivalent translational mass referred to the rack displacement x, and (b) the equivalent mass moment of inertia referred to pinion rotation θ.



Solution 9

1. Use the no-slip constraint
Choose compatible positive directions so that $x = R\theta$ and $\dot{x} = R\dot{\theta}$. Reversing one coordinate changes the sign, but not the squared terms in kinetic energy.

2. Write the total kinetic energy
The rack translates and the pinion rotates about its fixed center:

$$T = \tfrac{1}{2} m\dot{x}^2 + \tfrac{1}{2} J_0 \dot{\theta}^2.$$

3. Refer the energy to x
Use $\dot{\theta} = \dot{x}/R$:

$$T = \tfrac{1}{2}\left(m + \frac{J_0}{R^2}\right)\dot{x}^2 = \tfrac{1}{2} m_{\text{eq}} \dot{x}^2, \qquad \boxed{m_{\text{eq}} = m + J_0/R^2}.$$

J_0/R^2 has units kg, so it can be added to m.

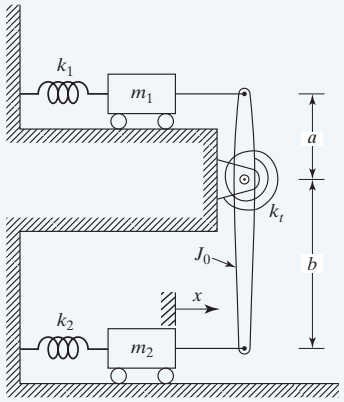
4. Refer the energy to θ
Instead use $\dot{x} = R\dot{\theta}$:

$$T = \tfrac{1}{2}(mR^2 + J_0)\dot{\theta}^2, \qquad \boxed{J_{\text{eq}} = J_0 + mR^2}.$$

Both terms now have units $\text{kg}\,\text{m}^2$. The relation $J_{\text{eq}} = R^2 m_{\text{eq}}$ confirms consistency between the two reference coordinates.

Question 10 | Rocker-arm equivalent mass — Application

For the rocker arm assembly, find the *equivalent mass* with respect to the displacement x of the lower slider (small motions). The upper slider has mass m_1, the lower slider has mass m_2; the rocker has mass moment of inertia J_0 about its pivot. The distances from the pivot to the upper and lower pins are a and b, respectively. (Springs k_1, k_2 and the torsional spring k_t affect stiffness, not mass.)



Solution 10

1. Relate the motions

Take lower-slider displacement x positive to the right. At small rocker angles, $x = b\theta$. With the upper-slider displacement also positive to the right, the two sliders move oppositely:

$$\dot{\theta} = \dot{x}/b, \qquad \dot{x}_1 = -\frac{a}{b}\dot{x}.$$

The minus sign matters for the physical motion; it disappears from kinetic energy.

2. Include both sliders and the rocker

Because J_0 is already the rocker's inertia about its fixed pivot,

$$T = \tfrac{1}{2}m_2\dot{x}^2 + \tfrac{1}{2}m_1\dot{x}_1^2 + \tfrac{1}{2}J_0\dot{\theta}^2.$$

Do not add a separate translational term for the rocker's center of mass when using its inertia about this fixed pivot.

3. Express all terms in the lower-slider coordinate

$$T = \tfrac{1}{2}\left[m_2 + m_1(a/b)^2 + J_0/b^2\right]\dot{x}^2.$$

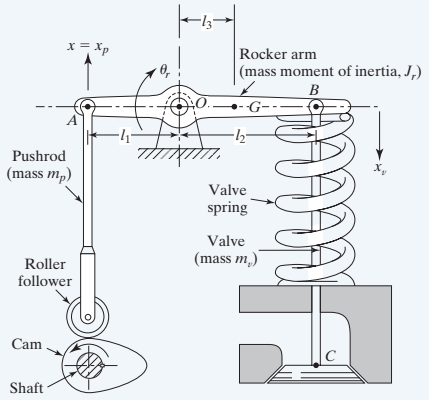
Matching $T = \frac{1}{2}m_{\text{eq}}\dot{x}^2$ gives

$$\boxed{m_{\text{eq}} = m_2 + m_1(a/b)^2 + J_0/b^2}.$$

Every term has units kg. Springs contribute potential energy and therefore stiffness; their stiffnesses do not enter this equivalent mass. The result assumes the sliders and rocker have the stated rigid kinematic connection.

Question 11 | Cam follower: two reference points — Challenge

A cam–follower drives a pushrod of mass m_p that acts on a rocker arm of mass m_r (with mass moment of inertia J_r about its C.G.). The valve has mass m_v. Let the distances from the rocker pivot to the pushrod joint, valve tip, and rocker C.G. be $l_1,\ l_2,\ l_3$, respectively. For small motions, find the equivalent mass m_{eq} of the system when referred to (i) point A (pushrod end) and (ii) point C (valve).



Solution 11

1. Choose pushrod displacement as the reference

Let $x = x_p$ at A. For a small rocker rotation, the magnitudes of the velocities are

$$|\dot{\theta}_r| = |\dot{x}|/l_1, \qquad |\dot{x}_v| = (l_2/l_1)|\dot{x}|, \qquad |v_G| = (l_3/l_1)|\dot{x}|.$$

The pushrod and valve can move in opposite directions; energy uses their squared speeds.

2. Write the rocker's energy correctly

Here J_r is about the center of mass, not the pivot. Include translation of the rocker center and rotation about it:

$$T = \tfrac{1}{2}m_p\dot{x}_p^2 + \tfrac{1}{2}m_v\dot{x}_v^2 + \tfrac{1}{2}m_r v_G^2 + \tfrac{1}{2}J_r\dot{\theta}_r^2.$$

Equivalently the rocker alone has inertia $J_O = J_r + m_r l_3^2$ about its fixed pivot.

3. Equivalent mass at A

Substitute all velocity ratios and equate to $\frac{1}{2}m_{\text{eq},A}\dot{x}^2$:

$$\boxed{m_{\text{eq},A} = m_p + m_v(l_2/l_1)^2 + \frac{J_r + m_r l_3^2}{l_1^2}}.$$

4. Equivalent mass at C

Use valve displacement as the reference. Then the corresponding speed magnitudes are $|\dot{x}_p| = (l_1/l_2)|\dot{x}_v|$, $|v_G| = (l_3/l_2)|\dot{x}_v|$ and $|\dot{\theta}_r| = |\dot{x}_v|/l_2$. Hence

$$\boxed{m_{\text{eq},C} = m_v + m_p(l_1/l_2)^2 + \frac{J_r + m_r l_3^2}{l_2^2}}.$$

The check $m_{\text{eq},A} = m_{\text{eq},C}(l_2/l_1)^2$ follows directly from equal kinetic energy. Spring mass is neglected.

Question 12 | Three viscous dampers — Foundation

Find a single equivalent damping constant for three linear viscous dampers of constants c_1, c_2, c_3 connected (a) in parallel and (b) in series.

Solution 12

1. Use the force–relative-velocity law

For each damper, the transmitted force magnitude is $F_i = c_i v_i$, where v_i is the relative velocity of its two ends. The force on a body opposes that relative motion.

2. Parallel connection

All dampers have the same relative velocity v. Their forces add:

$$F = c_1 v + c_2 v + c_3 v = c_{\text{eq}} v, \qquad \boxed{c_{\text{eq}} = c_1 + c_2 + c_3}.$$

3. Series connection

With massless intermediate connections, each damper transmits the same force F. The terminal relative velocity is the sum of the individual relative velocities:

$$v = v_1 + v_2 + v_3 = F\left(\frac{1}{c_1} + \frac{1}{c_2} + \frac{1}{c_3}\right).$$

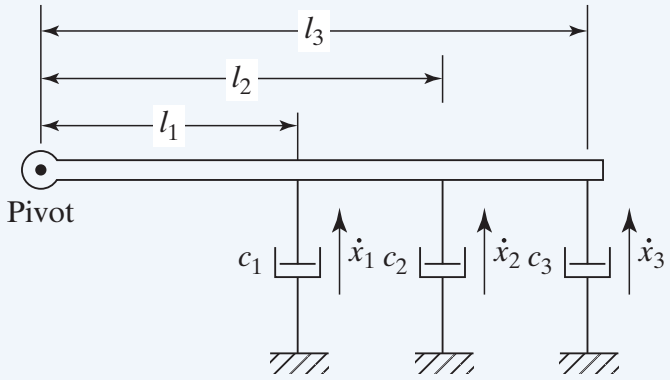
Therefore

$$\boxed{\frac{1}{c_{\text{eq}}} = \frac{1}{c_1} + \frac{1}{c_2} + \frac{1}{c_3}}.$$

These constants have units N s/m. The series formula follows from velocity compatibility, not from equality of damper displacements. A damper dissipates energy instead of storing it.

Question 13 | Dampers attached to a rigid bar — Application

Three viscous dampers of constants c_1, c_2, c_3 connect a massless rigid bar to fixed supports, at distances l_1, l_2, l_3 from its pivot. For small motion, find the equivalent damping constant referred to the displacement x_1 at the first damper.



Solution 13

1. Relate the attachment velocities

For a small rotation θ about the fixed pivot,

$$x_i = l_i\theta, \qquad \dot{x}_2 = (l_2/l_1)\dot{x}_1, \qquad \dot{x}_3 = (l_3/l_1)\dot{x}_1.$$

Because the other ends are fixed, these are also the relative velocities across the dampers.

2. Add the rates of energy dissipation

A linear viscous damper removes power $P_i = c_i\dot{x}_i^2$. Thus

$$P_{\text{loss}} = \left[c_1 + c_2(l_2/l_1)^2 + c_3(l_3/l_1)^2\right]\dot{x}_1^2.$$

An equivalent damper at the first attachment removes power $c_{\text{eq}}\dot{x}_1^2$. Equality gives

$$\boxed{c_{\text{eq}} = c_1 + c_2(l_2/l_1)^2 + c_3(l_3/l_1)^2}.$$

3. Check with a moment balance

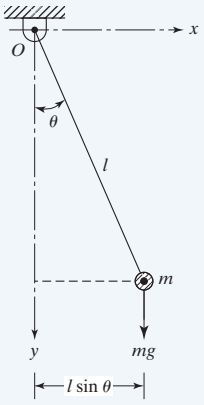
If force F acts at the first attachment, moment balance gives

$$Fl_1 = \sum_i (c_i\dot{x}_i)l_i = \frac{\dot{x}_1}{l_1}\sum_i c_i l_i^2.$$

Dividing by l_1 recovers $F = c_{\text{eq}}\dot{x}_1$. The squared ratios account for both the larger local velocity and its larger moment arm.

Question 14 | Pendulum equation and string tension — Application

A simple pendulum consists of a bob of mass m attached to a massless, inextensible string of length l. Let $\theta(t)$ be the angular displacement from the downward vertical. Derive the differential equation of motion for $\theta(t)$ and an expression for the string tension $T(t)$.



Solution 14

1. Define the coordinate and model

The bob moves on a circle of radius l. Assume the string remains taut, air resistance is negligible, and the pivot is fixed. Positive θ is measured from the downward vertical.

2. Take moments about the pivot

The tension has no moment about the pivot. Gravity produces restoring moment $-mgl\sin\theta$, while the bob's moment of inertia is ml^2. Thus

$$ml^2\ddot{\theta} = -mgl\sin\theta, \qquad \boxed{\ddot{\theta} + \frac{g}{l}\sin\theta = 0}.$$

This equation is nonlinear and applies beyond the small-angle approximation while the string remains taut.

3. Use radial force balance to find the tension

The speed is $v = l|\dot{\theta}|$ and the inward acceleration is $v^2/l = l\dot{\theta}^2$. Resolving forces toward the pivot gives

$$T - mg\cos\theta = ml\dot{\theta}^2, \qquad \boxed{T = mg\cos\theta + ml\dot{\theta}^2}.$$

If a computed tension is negative, a string cannot supply it: the assumed taut circular motion is no longer valid.

4. Connect the exact model to small motion and energy

For $|\theta| \ll 1$ rad, $\sin\theta \simeq \theta$, so

$$ml^2\ddot{\theta} + mgl\theta = 0.$$

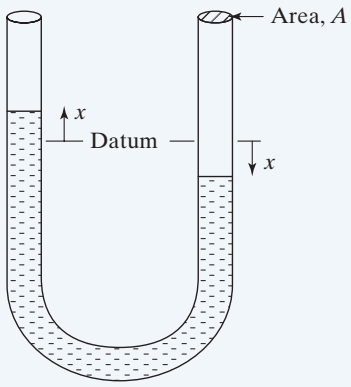
Measured from the lowest position, the energies are

$$T_{\text{kin}} = \tfrac{1}{2}ml^2\dot{\theta}^2, \qquad U = mgl(1-\cos\theta) \simeq \tfrac{1}{2}mgl\theta^2.$$

Thus the equivalent inertia is ml^2 and the small-motion torsional stiffness is mgl. Here T_{kin} denotes kinetic energy, whereas T above denotes tension. Solving the free-vibration equation is a later step.

Question 15 | Restoring stiffness of a U-tube — Application

A U-tube manometer, open at both ends, contains a liquid column of total length l and cross-sectional area A. The liquid has specific weight γ. For a small displacement x of one meniscus from the datum (the other meniscus moves by $-x$), determine the *equivalent spring constant* associated with the restoring force.



Solution 15

1. Use volume conservation

The two limbs have the same area A. If one surface rises by x, the other falls by x, so their vertical separation is $2x$.

2. Find the unbalanced hydrostatic head

The corresponding hydrostatic pressure imbalance at a common level is

$$\Delta p = \gamma(2x).$$

Both free surfaces are at atmospheric pressure; Δp is not a pressure difference between those free surfaces. It is the head imbalance that drives fluid from the higher side toward the lower side.

3. Identify the restoring force

Multiplying by the cross-sectional area gives the force magnitude $2\gamma Ax$. If positive x means a rise in the chosen limb, the force opposing that rise is

$$F_{\text{rest}} = -2\gamma Ax = -k_{\text{eq}}x, \qquad \boxed{k_{\text{eq}} = 2\gamma A}.$$

Since $[\gamma] = \text{N/m}^3$, the stiffness has units N/m.

4. Relate stiffness and inertia to the coordinate

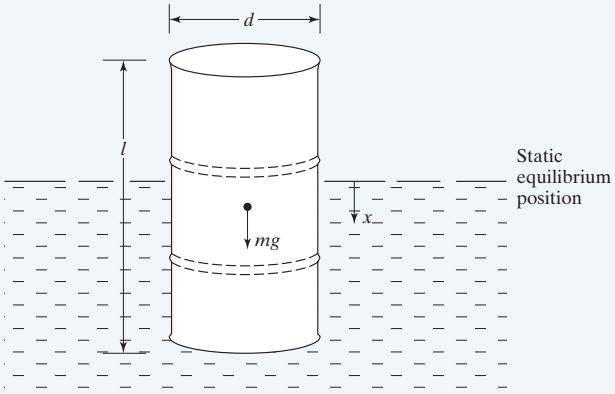
The total liquid mass is $m_{\text{eq}} = \rho Al = (\gamma/g)Al$. In a uniform tube, each fluid element has the same speed magnitude $|\dot{x}|$, so

$$T_{\text{kin}} = \tfrac{1}{2}\rho Al\dot{x}^2, \qquad U = \tfrac{1}{2}k_{\text{eq}}x^2 = \gamma Ax^2.$$

Neglecting flow losses, the model is $\rho Al\ddot{x} + 2\gamma Ax = 0$. Liquid length affects inertia, but not the restoring stiffness. The approximation requires surfaces to remain in the uniform vertical limbs.

Question 16 | Buoyant restoring stiffness — Application

An oil drum of diameter d and mass m floats in sea water of density ρ_w. For a small vertical displacement x of the drum from its static equilibrium position, determine the *equivalent spring constant* associated with the restoring (buoyant) force.



Solution 16

1. Find the equilibrium displacement of water

Let V_0 be the submerged volume at equilibrium and $A = \pi d^2/4$ the constant waterplane area. Archimedes' principle gives

$$\rho_w g V_0 = mg.$$

2. Displace the drum downward by x

For vertical sides, and while the waterline remains on those sides, the additional displaced volume is Ax. The new upward buoyancy is

$$B = \rho_w g(V_0 + Ax) = mg + \rho_w g A x.$$

3. Subtract the equilibrium forces

With downward positive, the net force is

$$F_{\text{rest}} = mg - B = -\rho_w g A x.$$

Comparing with $F_{\text{rest}} = -k_{\text{eq}} x$ gives

$$\boxed{k_{\text{eq}} = \rho_w g A = \rho_w g \frac{\pi d^2}{4}}.$$

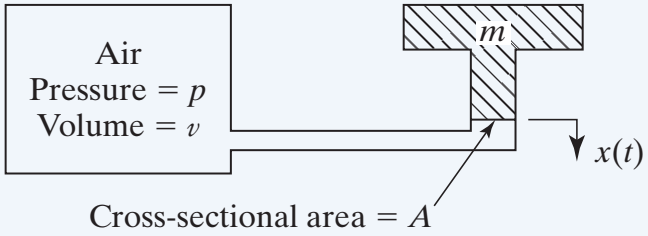
The sign shows that a downward displacement produces an upward restoring force. Units are $(\text{kg/m}^3)(\text{m/s}^2)(\text{m}^2) = \text{N/m}$.

4. State the limits of the model

The drum mass determines the equilibrium draft but does not appear explicitly in this stiffness for a constant circular waterplane. Assume pure vertical motion without tilt, small displacement, and no complete submergence. Hydrodynamic added mass and fluid damping would require further modelling.

Question 17 | Linearisation of an air spring — Challenge

An air spring consists of a chamber of volume v connected to a piston of area A supporting a mass m. Assuming the gas changes *adiabatically* ($p\,v^{\gamma} = \text{const}$ with $\gamma = 1.4$ for air), find the equivalent spring constant relating the piston displacement x to the restoring force.



Solution 17

1. Define the equilibrium state

Let p_0, v_0 denote the equilibrium *absolute* gas pressure and volume. Take piston displacement x positive downward, so that compression gives

$$v(x) = v_0 - Ax.$$

Assume a constant piston area, negligible leakage/friction, and constant external pressure. Static gas force, weight and external pressure balance at $x = 0$.

2. Use the adiabatic pressure–volume relation

Since $pv^{\gamma} = p_0 v_0^{\gamma}$,

$$p(x) = p_0 \left(\frac{v_0}{v_0 - Ax} \right)^{\gamma}.$$

This is a nonlinear force relation, even though the piston area is constant.

3. Differentiate at equilibrium

Either differentiating the expression above or using $dp/dv = -\gamma p/v$ gives

$$\left.\frac{dp}{dx}\right|_0 = \left.\frac{dp}{dv}\frac{dv}{dx}\right|_0 = \left(-\gamma \frac{p_0}{v_0}\right)(-A) = \gamma \frac{p_0 A}{v_0}.$$

4. Find the incremental restoring force

The increase in upward gas force is $A[p(x) - p_0]$. Therefore the net force, downward positive, is

$$F_{\text{rest}} \simeq -A \left.\frac{dp}{dx}\right|_0 x = -\frac{\gamma p_0 A^2}{v_0} x.$$

Hence

$$\boxed{k_{\text{eq}} = \frac{\gamma p_0 A^2}{v_0} = \frac{1.4 p_0 A^2}{v_0}}.$$

Its units are $(\text{N/m}^2)(\text{m}^4)/\text{m}^3 = \text{N/m}$. The linearisation requires $|Ax| \ll v_0$. Use absolute pressure in the gas law; a gauge pressure cannot simply replace p_0.

Question 18 | Harmonic-motion terminology — Foundation

(a) Define cycle, amplitude, phase angle, frequency, period, and natural frequency.

(b) State the relations between period, angular frequency, and frequency in cycles per second.

(c) Explain how the frequency, phase, and amplitude of a harmonic motion are obtained from its rotating-vector representation.

Solution 18

1. Define the terms using $x = A\cos(\omega t + \phi)$

- **Cycle:** one complete repetition of the motion, returning to the same displacement and direction of motion.
- **Amplitude** A**:** the largest magnitude of displacement from the mean position. Peak-to-peak displacement is $2A$.
- **Phase angle:** $\omega t + \phi$ specifies the location within the cycle; ϕ is the phase at the chosen time origin.
- **Period** T**:** time for one cycle, in seconds.
- **Frequency** f**:** cycles per second, in hertz (Hz).
- **Natural frequency:** a frequency at which a system can vibrate freely after an initial disturbance. It depends on the system properties and model, rather than an arbitrarily imposed drive speed.

2. Relate the frequency measures

One cycle corresponds to 2π radians, so $\omega T = 2\pi$. Hence

$$\boxed{f = \frac{1}{T}, \qquad \omega = 2\pi f = \frac{2\pi}{T}}.$$

ω is in rad/s, whereas f is in Hz. They describe the same repetition rate in different units.

3. Interpret the rotating vector

A vector of length A rotating uniformly at angular speed ω has initial angle ϕ. Its projection on the horizontal axis is $A\cos(\omega t + \phi)$. Thus vector length gives amplitude, angular speed gives angular frequency, and orientation at $t = 0$ gives phase. A complete revolution takes $T = 2\pi/\omega$.

Question 19 | Rotational speed and period **Foundation**

(a) Convert a rotational speed of 1000 rpm to radians per second.

(b) When a turbine rotates at 6000 rpm, how long does it take to complete one revolution?

Solution 19

1. Convert revolutions per minute to revolutions per second
For a speed N rpm,

$$f = \frac{N}{60} \text{ Hz}.$$

2. Apply the conversion to the first speed
At 1000 rpm, $f = 1000/60 = 16.6667$ Hz. Multiplying by 2π radians per revolution gives

$$\boxed{\omega = \frac{1000(2\pi)}{60} = 104.720 \text{ rad/s}}.$$

3. Find the period at the second speed
At 6000 rpm, $f = 6000/60 = 100$ revolutions/s. Therefore

$$\boxed{T = \frac{1}{f} = \frac{1}{100} = 0.0100 \text{ s} = 10.0 \text{ ms}}.$$

The two parts use different given speeds. A revolution is a complete cycle; one radian is only $1/(2\pi)$ of a revolution.

Question 20 | Period and peak velocity/acceleration **Foundation**

A harmonic motion has an amplitude of 0.05 m and a frequency of 10 Hz. Find its period, maximum velocity, and maximum acceleration.

Solution 20

1. Find the period and angular frequency

$$T = \frac{1}{10} = \boxed{0.100 \text{ s}}, \qquad \omega = 2\pi(10) = 62.8319 \text{ rad/s}.$$

2. Differentiate a general harmonic displacement
For any initial phase ϕ,

$$x = A\cos(\omega t + \phi), \quad \dot{x} = -\omega A \sin(\omega t + \phi), \quad \ddot{x} = -\omega^2 A \cos(\omega t + \phi).$$

Because the largest absolute value of sine or cosine is one, the peak speed is ωA and the peak acceleration magnitude is $\omega^2 A$.

3. Substitute the data

$$\boxed{v_{\max} = 62.8319(0.05) = 3.14159 \text{ m/s}},$$

$$\boxed{a_{\max} = (62.8319)^2(0.05) = 197.392 \text{ m/s}^2}.$$

The acceleration peak occurs at a displacement extreme, where velocity is zero. The speed peak occurs as the motion passes through its mean position. Frequency in Hz must first be converted to angular frequency before using these amplitude formulas.

Question 21 | Velocity and acceleration histories **Application**

The displacement of a machine is

$$x(t) = 0.2\sin(5t + 3),$$

where x is in meters and t in seconds. Find its velocity and acceleration as functions of time, and the amplitudes of displacement, velocity, and acceleration.

Solution 21

1. Identify the parameters
Here $A = 0.2$ m, $\omega = 5$ rad/s, and the initial phase is 3 rad. The phase changes when the time origin changes; the amplitude does not.
2. Differentiate once
Using the chain rule,

$$v(t) = \frac{dx}{dt} = 0.2(5)\cos(5t + 3) = \boxed{1.0\cos(5t + 3)\ \text{m/s}}.$$

The factor 5 comes from differentiating the argument $5t + 3$.
3. Differentiate again

$$a(t) = \frac{dv}{dt} = -1.0(5)\sin(5t + 3) = \boxed{-5.0\sin(5t + 3)\ \text{m/s}^2}.$$

Thus $a(t) = -25x(t) = -\omega^2 x(t)$, consistent with harmonic motion.
4. Read the amplitudes and phases

$$\boxed{X = 0.2\ \text{m}, \qquad V = 1.0\ \text{m/s}, \qquad A_a = 5.0\ \text{m/s}^2}.$$

Since $\cos\psi = \sin(\psi + \pi/2)$, velocity leads displacement by $\pi/2$. Since $-\sin\psi = \sin(\psi + \pi)$, acceleration leads displacement by π. The minus sign is a phase/direction change, not a negative amplitude.

Question 22 | Amplitude and phase from initial data — Challenge

A machine is subjected to the motion

$$x(t) = A\cos(50t + \alpha) \text{ mm},$$

with initial conditions $x(0) = 3$ mm and $\dot{x}(0) = 1.0$ m/s.
(a) Find the constants A and α. (b) Express the motion in the form $x(t) = A_1\cos(\omega t) + A_2\sin(\omega t)$ and identify A_1, A_2.

Solution 22

1. Use consistent displacement units
The stated displacement is in millimeters, so convert the initial velocity:

$$\dot{x}(0) = 1.0 \text{ m/s} = 1000 \text{ mm/s}.$$

The angular frequency is $\omega = 50$ rad/s.
2. Apply both initial conditions
At $t = 0$,

$$A\cos\alpha = 3.$$

Differentiate the displacement to get $\dot{x} = -50A\sin(50t + \alpha)$. Hence

$$-50A\sin\alpha = 1000, \qquad A\sin\alpha = -20.$$

3. Solve for amplitude and phase
Square and add the two component equations:

$$A^2(\cos^2\alpha + \sin^2\alpha) = 3^2 + (-20)^2 = 409.$$

Taking amplitude nonnegative gives

$$\boxed{A = \sqrt{409} = 20.22375 \text{ mm}}.$$

The cosine component is positive and the sine component negative, so α lies in quadrant IV:

$$\boxed{\alpha = \operatorname{atan2}(-20, 3) = -1.42191 \text{ rad}}$$

or any equivalent angle differing by 2π.
4. Write the sine–cosine form
Expand $A\cos(50t + \alpha)$:

$$x = (A\cos\alpha)\cos 50t - (A\sin\alpha)\sin 50t.$$

Thus

$$\boxed{x(t) = 3\cos 50t + 20\sin 50t \text{ mm}}, \qquad \boxed{A_1 = 3 \text{ mm}, \quad A_2 = 20 \text{ mm}}.$$

5. Check both conditions
At $t = 0$, $x = 3$ mm. Differentiation gives $\dot{x} = -150\sin 50t + 1000\cos 50t$ mm/s, so $\dot{x}(0) = 1000$ mm/s. Both original conditions are satisfied.

Question 23 | A complex number in polar form **Foundation**

Express the complex number $5 + 2i$ in exponential form $Ae^{i\theta}$.

Solution 23

1. Match real and imaginary parts

Euler's identity gives $Ae^{i\theta} = A\cos\theta + iA\sin\theta$. Therefore

$$A\cos\theta = 5, \qquad A\sin\theta = 2.$$

2. Find the magnitude

Squaring and adding gives $A^2 = 5^2 + 2^2 = 29$, so

$$\boxed{A = \sqrt{29} = 5.38516}.$$

3. Find the angle with the correct quadrant

Both components are positive, so the angle is in quadrant I:

$$\theta = \operatorname{atan2}(2, 5) = 0.380506 \text{ rad} = 21.8014^\circ.$$

Thus

$$\boxed{5 + 2i = \sqrt{29}\, e^{i\,0.380506}}.$$

The angle in the exponential is in radians. Substitution into the cosine and sine components returns 5 and 2. In a harmonic-motion phasor, this magnitude and angle would represent amplitude and phase.

Question 24 | Addition at a common frequency **Application**

Find the sum of the two harmonic motions

$$x_1(t) = 10\cos(\omega t), \qquad x_2(t) = 15\cos(\omega t + 2).$$

Solution 24

Both terms have the same angular frequency. The phase shift 2 is in radians; the amplitudes use the same displacement unit.

1. Expand the shifted cosine

$$15\cos(\omega t + 2) = 15\cos 2\cos\omega t - 15\sin 2\sin\omega t.$$

Consequently,

$$x = C\cos\omega t - S\sin\omega t,$$

where $C = 10 + 15\cos 2 = 3.757797$ and $S = 15\sin 2 = 13.639461$.

2. Match a single harmonic motion

Write $x = A\cos(\omega t + \alpha)$. Expanding this expression shows

$$A\cos\alpha = C, \qquad A\sin\alpha = S.$$

Therefore

$$A = \sqrt{C^2 + S^2} = 14.147648, \qquad \alpha = \operatorname{atan2}(S, C) = 1.301957 \text{ rad}.$$

The resultant is

$$\boxed{x(t) = 14.14765\cos(\omega t + 1.301957)}.$$

Its amplitude has the same displacement unit as the two given amplitudes.

3. Verify with complex amplitudes

The two phasors are $X_1 = 10$ and $X_2 = 15e^{2i}$. Their sum is

$$X = X_1 + X_2 = (10 + 15\cos 2) + i(15\sin 2) = C + iS.$$

Thus $|X| = A$ and $\arg X = \alpha$, giving the same result. At $t = 0$, the direct sum is $10 + 15\cos 2 = C$, and the resultant gives $A\cos\alpha = C$. Amplitudes cannot simply be added unless the motions are in phase.

Question 25 | Periodic does not always mean harmonic — Challenge

Consider two harmonic motions

$$x_1(t) = 2\cos(2t), \qquad x_2(t) = \cos(3t).$$

Is the sum $x(t) = x_1 + x_2$ a harmonic motion? If not, is it periodic? If periodic, what is its fundamental period?

Solution 25

1. Distinguish harmonic from periodic motion

A single harmonic motion has one angular frequency and can be written $A\cos(\omega t + \phi)$. The given sum contains two nonzero components with distinct angular frequencies, 2 and 3 rad/s. It cannot be expressed as one sinusoid with constant amplitude and phase.

2. Verify that it is not simple harmonic

For $x = 2\cos 2t + \cos 3t$,

$$\ddot{x} = -8\cos 2t - 9\cos 3t.$$

If $\ddot{x} = -\omega^2 x$ held for every t, comparison of the independent cosine terms would require both $\omega^2 = 4$ and $\omega^2 = 9$, which is impossible.

3. Find a common repetition time

The individual periods are

$$T_1 = \frac{2\pi}{2} = \pi \text{ s}, \qquad T_2 = \frac{2\pi}{3} \text{ s}.$$

A common period must satisfy $T = m\pi = n(2\pi/3)$ for positive integers m, n, or $3m = 2n$. The smallest solution is $m = 2, n = 3$, giving

$$\boxed{T = 2\pi \text{ s} \approx 6.28319 \text{ s}}.$$

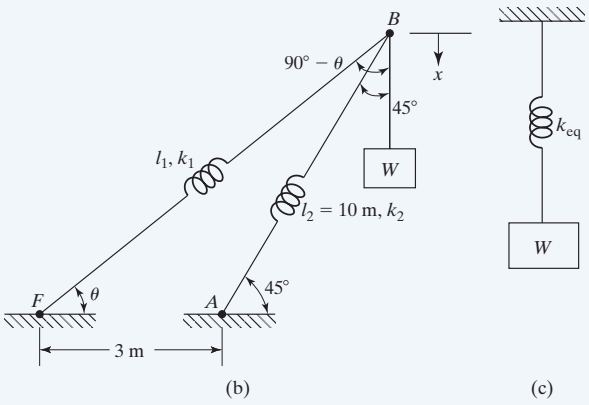
Because both frequency components are nonzero, this is the fundamental period. Equivalently the fundamental angular frequency is 1 rad/s, with components at its second and third harmonics.

4. State the conclusion

The motion is **periodic but not a single harmonic motion**. A fixed phasor addition at one common frequency cannot combine these two components. Their frequency ratio is rational, which allows a common period.

Question 26 | Vertical stiffness of a crane — Challenge

The boom AB of a crane is a uniform steel bar of length 10 m and area $A_2 = 2500$ mm^2. A steel cable FB of area $A_1 = 100$ mm^2 supports the boom. Neglect the effect of cable $CDEB$. Find the equivalent vertical spring constant at the hook, referred to point B, with $E = 207$ GPa for both members. Use the geometry shown and the one-coordinate model in which B moves vertically.



Solution 26

1. Find the cable length and direction

The supports F, A are 3 m apart; the boom is inclined at 45°. The angle FAB is 135°. The cosine rule gives

$$l_1 = \sqrt{3^2 + 10^2 - 2(3)(10)\cos 135^\circ} = 12.3055 \text{ m}.$$

For the cable angle θ measured from the horizontal at F,

$$\cos\theta = \frac{l_1^2 + 3^2 - 10^2}{2(3)l_1}, \qquad \theta = 35.0736^\circ.$$

2. Calculate the axial stiffnesses

Convert mm^2 to m^2 using 1 mm^2 $= 10^{-6}$ m^2:

$$k_1 = \frac{A_1 E}{l_1} = \frac{100(10^{-6})207(10^9)}{12.3055} = 1.6822 \times 10^6 \text{ N/m},$$

$$k_2 = \frac{A_2 E}{10} = 5.1750 \times 10^7 \text{ N/m}.$$

3. Project the prescribed vertical displacement

For a small downward displacement x, the magnitudes of the axial changes are $|\delta_1| = x\sin\theta$ and $|\delta_2| = x\cos 45^\circ$. A shortening and an extension both contribute positively to elastic energy:

$$U = \tfrac{1}{2}k_1\delta_1^2 + \tfrac{1}{2}k_2\delta_2^2 = \tfrac{1}{2}\left(k_1\sin^2\theta + k_2\cos^2 45^\circ\right)x^2.$$

4. Identify the equivalent spring

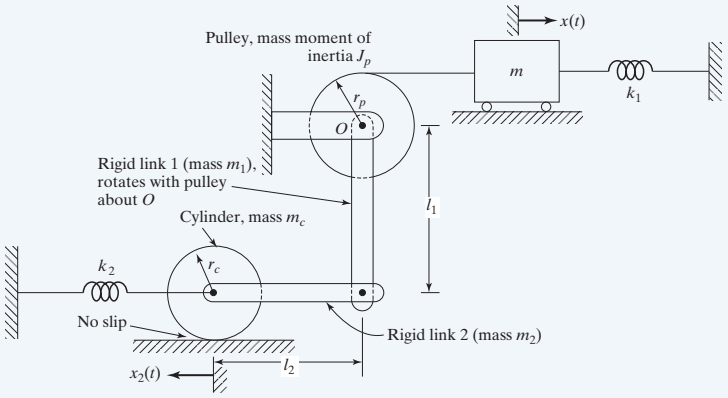
Equate this energy to $\frac{1}{2}k_{\text{eq}}x^2$:

$$\boxed{k_{\text{eq}} = k_1\sin^2\theta + k_2\cos^2 45^\circ = 2.64304 \times 10^7 \text{ N/m}.}$$

This is the stiffness for the stated vertical-motion idealisation. Allowing independent horizontal motion at B requires a coupled displacement calculation. Bending, cable slackening and stiffness changes due to initial load are neglected.

Question 27 | Pulley, links and a rolling cylinder — Challenge

Find the equivalent mass referred to $x(t)$ for the system shown. The pulley of radius r_p and inertia J_p is rigidly attached to uniform link 1 of length l_1 and mass m_1. Link 2 has mass m_2 and connects link 1 to a solid cylinder of radius r_c and mass m_c that rolls without slip. The block has mass m. Use small motions about the shown configuration and no slip between belt and pulley.



Solution 27

1. Express all speeds in terms of $\dot{x}$

The pulley and link 1 rotate together: $|\dot{\theta}_p| = |\dot{\theta}_1| = |\dot{x}|/r_p$. At the shown position, link 1 is vertical, so its lower joint moves horizontally to first order. Link 2 is horizontal and translates to first order. Thus

$$|\dot{x}_2| = \frac{l_1}{r_p}|\dot{x}|, \qquad |\dot{\theta}_c| = \frac{|\dot{x}_2|}{r_c} = \frac{l_1}{r_p r_c}|\dot{x}|.$$

2. Include every body's kinetic energy

Let J_1 be link 1's inertia about O and J_c the cylinder's inertia about its centre. Then

$$T = \tfrac{1}{2}m\dot{x}^2 + \tfrac{1}{2}J_p\dot{\theta}_p^2 + \tfrac{1}{2}J_1\dot{\theta}_1^2 + \tfrac{1}{2}m_2\dot{x}_2^2 + \tfrac{1}{2}m_c\dot{x}_2^2 + \tfrac{1}{2}J_c\dot{\theta}_c^2.$$

The cylinder both translates and rotates. Link 1's inertia about O already includes its centre-of-mass translation.

3. Collect the coefficient of $\dot{x}^2/2$

Substituting the speed ratios and equating $T = \tfrac{1}{2}m_{\text{eq}}\dot{x}^2$ gives

$$\boxed{m_{\text{eq}} = m + \frac{J_p + J_1}{r_p^2} + \left(m_2 + m_c + \frac{J_c}{r_c^2}\right)\left(\frac{l_1}{r_p}\right)^2.}$$

4. Use the stated body shapes

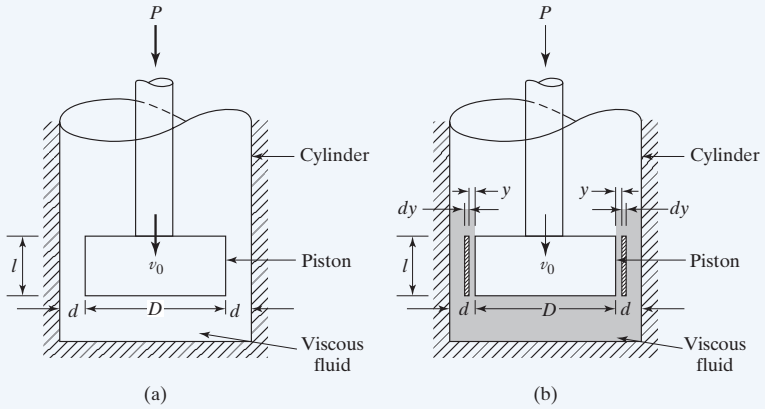
For a uniform slender link, $J_1 = m_1 l_1^2/3$; for a solid cylinder, $J_c = m_c r_c^2/2$. Hence

$$\boxed{m_{\text{eq}} = m + \frac{J_p}{r_p^2} + \left(\frac{m_1}{3} + m_2 + \frac{3}{2}m_c\right)\left(\frac{l_1}{r_p}\right)^2.}$$

Each term has units kg. The factor 3/2 contains the cylinder's translation and rolling rotation. Springs affect stiffness, not this equivalent mass.

Question 28 | Damping of a piston-cylinder dashpot — Challenge

Develop an expression for the damping constant of the piston-cylinder dashpot. A piston of diameter D and length l moves at speed v_0 through a Newtonian fluid of viscosity μ. Its uniform radial clearance from the cylinder wall is $d \ll D$. Assume slow, laminar, incompressible flow through the clearance.



Solution 28

1. Choose the flow coordinate and pressure force

Take fluid velocity $v(y)$ positive upward through the gap, opposite to the downward piston motion. Measure y outward from the piston: $0 \le y \le d$. No slip gives

$$v(0) = -v_0, \qquad v(d) = 0.$$

Let Δp be the pressure difference between the piston faces and $A_p = \pi D^2/4$. In the thin-gap, pressure-dominated model, the resisting force magnitude is $P \simeq A_p \Delta p$.

2. Obtain the velocity profile

Approximating the annulus as a thin planar gap of length l, the steady axial force balance is

$$\mu \frac{d^2 v}{dy^2} = -\frac{\Delta p}{l}.$$

Integrate twice:

$$v(y) = -\frac{\Delta p}{2\mu l} y^2 + C_1 y + C_2.$$

The first boundary condition gives $C_2 = -v_0$. The second gives $C_1 = \Delta p\, d/(2\mu l) + v_0/d$. Therefore

$$\boxed{v(y) = \frac{\Delta p}{2\mu l} y(d - y) - v_0 \left(1 - \frac{y}{d}\right).}$$

The first term is pressure-driven return flow; the second is fluid dragged downward by the piston.

3. Integrate the return flow

The thin annulus has circumference approximately πD. Thus

$$Q = \pi D \int_0^d v(y)\, dy.$$

Since $\int_0^d y(d - y)\, dy = d^3/6$ and $\int_0^d (1 - y/d)\, dy = d/2$,

$$Q = \pi D \left(\frac{\Delta p\, d^3}{12\mu l} - \frac{v_0 d}{2} \right).$$

Solution 28 (continued)

4. Apply volume conservation

The displaced piston volume returns through the gap, so $Q = A_p v_0$. Dividing by πD gives

$$\frac{D}{4} v_0 = \frac{\Delta p \, d^3}{12 \mu l} - \frac{d}{2} v_0.$$

Rearrange to obtain

$$\Delta p = \frac{12 \mu l}{d^3} \left(\frac{D}{4} + \frac{d}{2} \right) v_0 = \frac{3 \mu l D}{d^3} \left(1 + \frac{2d}{D} \right) v_0.$$

5. Identify damping and its approximation

Using $P \simeq A_p \Delta p = c \, v_0$ gives the textbook thin-gap result

$$\boxed{c \simeq \frac{3 \pi \mu D^3 l}{4 d^3} \left(1 + \frac{2d}{D} \right).}$$

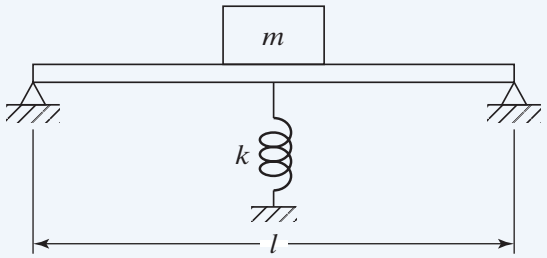
For $d/D \ll 1$, the leading approximation is

$$\boxed{c \simeq \frac{3 \pi \mu D^3 l}{4 d^3}} \qquad [\mathrm{N\,s/m}].$$

The $2d/D$ term comes from piston-driven flow in the clearance, not an end-effect correction. The model neglects entrance losses, annular curvature corrections and the smaller direct side-shear contribution to the piston force; it is not an exact finite-clearance formula. The force on the moving piston opposes its velocity.

Question 29 | Reducing a beam-supported machine deflection — Application

A machine of mass $m = 500$ kg is mounted at mid-span on a simply supported steel beam of length $l = 2$ m. Its rectangular section has depth $h = 0.1$ m and width $b = 1.2$ m; $E = 2.06 \times 10^{11}$ N/m^2. A vertical spring of stiffness k is added beneath the mid-span. Find k to reduce the original deflection by (a) 25%, (b) 50%, and (c) 75%. Neglect beam mass.



Solution 29

1. Replace the beam by its mid-span stiffness

For a central force W on a simply supported beam,

$$\delta_0 = \frac{Wl^3}{48EI}, \qquad I = \frac{bh^3}{12} = \frac{1.2(0.1)^3}{12} = 1.00 \times 10^{-4}\ \text{m}^4.$$

Consequently,

$$k_b = \frac{W}{\delta_0} = \frac{48EI}{l^3} = \frac{48(2.06 \times 10^{11})(10^{-4})}{2^3} = 1.236 \times 10^8\ \text{N/m}.$$

2. Combine the beam and added spring

The beam and spring share the same mid-span deflection; their upward forces add. Assuming the added spring is uncompressed at the beam's undeformed position,

$$W = (k_b + k)\delta, \qquad \delta = \frac{W}{k_b + k}.$$

3. Translate a reduction into remaining deflection

Let r be the fractional reduction, so $\delta = (1 - r)\delta_0$. Then

$$\frac{\delta}{\delta_0} = \frac{k_b}{k_b + k} = 1 - r \quad \Longrightarrow \quad \boxed{k = k_b \frac{r}{1 - r}.}$$

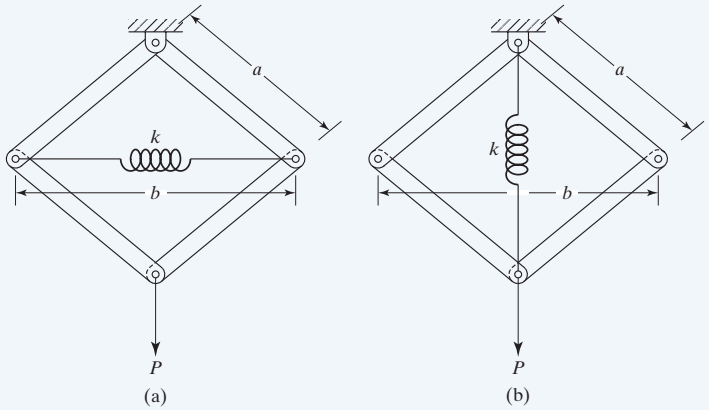
4. Evaluate all three cases

Reduction r	δ/δ_0	k (N/m)
0.25	0.75	$(1.236 \times 10^8)/3 = 4.12 \times 10^7$
0.50	0.50	1.236×10^8
0.75	0.25	$3(1.236 \times 10^8) = 3.708 \times 10^8$

With $g = 9.81$ m/s^2, $W = 4905$ N and $\delta_0 = 3.9684 \times 10^{-5}$ m. The new deflections are respectively $0.75\delta_0$, $0.50\delta_0$ and $0.25\delta_0$. The load cancels from the required stiffness ratios.

Question 30 | A rhombus with a horizontal or vertical spring — Challenge

Four identical rigid bars of length a form a rhombus whose top joint is fixed. The horizontal distance between its side joints is b. A load P acts at the bottom joint. A spring of stiffness k joins (a) the side joints horizontally or (b) the top and bottom joints vertically. Find the equivalent stiffness for small vertical displacement x of the bottom joint. Neglect masses and joint friction, and take the spring as unstrained in the reference configuration.



Solution 30

1. Relate height and width

Let H be the vertical distance between the top and bottom joints. Each quarter of the rhombus is a right triangle:

$$\left(\frac{H}{2}\right)^2 + \left(\frac{b}{2}\right)^2 = a^2, \qquad H^2 + b^2 = 4a^2.$$

Differentiate with fixed a: $H\,dH + b\,db = 0$. With $x = \Delta H$,

$$\Delta b \simeq -\frac{H}{b}\,x.$$

A downward bottom displacement increases height and decreases width.

2. Horizontal spring

The incremental spring deformation is Δb. Its sign disappears in energy:

$$U = \tfrac{1}{2}k(\Delta b)^2 \simeq \tfrac{1}{2}k\left(\frac{H}{b}\right)^2 x^2.$$

Equating this to $\frac{1}{2}k_{\text{eq}}^{(a)}x^2$ gives

$$\boxed{k_{\text{eq}}^{(a)} = k\frac{H^2}{b^2} = k\frac{4a^2 - b^2}{b^2}.}$$

3. Vertical spring

Its deformation equals the bottom displacement, so $U = \frac{1}{2}kx^2$ and

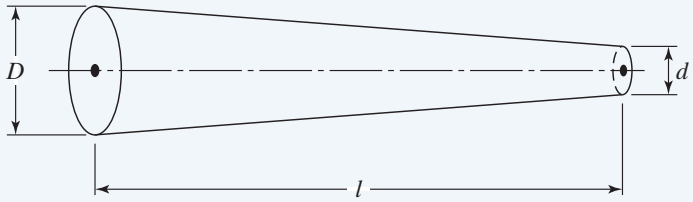
$$\boxed{k_{\text{eq}}^{(b)} = k.}$$

4. Interpret the geometry and limits

For a flatter rhombus, b increases and H decreases, so the horizontal-spring contribution to vertical stiffness *decreases*. The vertical spring acts directly. These are local, small-displacement results for $0 < b < 2a$, away from a collapsed linkage. A preloaded horizontal spring would add a geometric-stiffness term and requires its initial force to be specified.

Question 31 | An equivalent hollow shaft for a tapered bar — Challenge

Find the length L_{eq} of a uniform hollow circular shaft, of inner diameter d and wall thickness t, having the same axial spring constant as the solid conical shaft shown. The conical shaft has length l and end diameters D and d. Both shafts have the same Young's modulus E.



Solution 31

1. Describe the varying cross section

Measure s from the small end. The local diameter and area are

$$d_s = d + \frac{D-d}{l}s, \qquad A(s) = \frac{\pi}{4}d_s^2.$$

Under an axial force F, every section transmits the same force. An element of length ds extends by $d\delta = F\,ds/[EA(s)]$.

2. Integrate the axial compliance

Thus

$$\frac{\delta}{F} = \frac{1}{k_{\text{cone}}} = \frac{4}{E\pi}\int_0^l \frac{ds}{[d+(D-d)s/l]^2}.$$

Use $u = d + (D-d)s/l$, so $ds = l\,du/(D-d)$:

$$\begin{aligned}\frac{1}{k_{\text{cone}}} &= \frac{4l}{E\pi(D-d)}\int_d^D u^{-2}\,du \\ &= \frac{4l}{E\pi(D-d)}\left(\frac{1}{d}-\frac{1}{D}\right) = \frac{4l}{E\pi dD}.\end{aligned}$$

Therefore $k_{\text{cone}} = E\pi dD/(4l)$. For $D = d$, the same result follows by the uniform-bar limit.

3. Find the hollow-shaft area

Its outer diameter is $d + 2t$, hence

$$A_h = \frac{\pi}{4}[(d+2t)^2 - d^2] = \pi(dt + t^2), \qquad k_h = \frac{EA_h}{L_{\text{eq}}}.$$

4. Match stiffnesses and solve for length

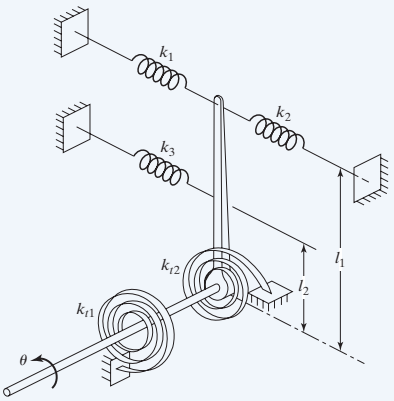
Set $k_h = k_{\text{cone}}$:

$$\frac{E\pi(dt+t^2)}{L_{\text{eq}}} = \frac{E\pi dD}{4l} \quad\Longrightarrow\quad \boxed{L_{\text{eq}} = \frac{4l(dt+t^2)}{dD}}.$$

Young's modulus cancels because the materials are identical. The area ratio is dimensionless, so the result has units of length. The calculation concerns axial deformation; it is not a torsional-stiffness equivalence.

Question 32 | Combining linear and torsional springs — Application

A rigid vertical bar is fixed to a shaft that rotates by a small angle θ. Springs k_1 and k_2 connect to the bar at distance l_1 from the shaft axis, on opposite sides; spring k_3 connects at distance l_2. Two torsional springs k_{t1} and k_{t2} also oppose the shaft rotation. Find the equivalent spring constant referred to θ.



Solution 32

1. Relate each linear deformation to rotation

The shaft and bar are rigid, so both torsional springs rotate through θ. At small angles, the bar attachment displacements have magnitudes

$$|x_1| = |x_2| \simeq l_1|\theta|, \qquad |x_3| \simeq l_2|\theta|.$$

One of the opposing upper springs lengthens while the other shortens. Both oppose the rotation.

2. Add all elastic-energy contributions

Linear springs store energy $kx^2/2$; torsional springs store $k_t\theta^2/2$. Consequently,

$$\begin{aligned} U &= \tfrac{1}{2}k_1(l_1\theta)^2 + \tfrac{1}{2}k_2(l_1\theta)^2 + \tfrac{1}{2}k_3(l_2\theta)^2 \\ &\quad + \tfrac{1}{2}k_{t1}\theta^2 + \tfrac{1}{2}k_{t2}\theta^2 \\ &= \tfrac{1}{2}\left[k_{t1} + k_{t2} + (k_1 + k_2)l_1^2 + k_3 l_2^2\right]\theta^2. \end{aligned}$$

3. Identify the equivalent torsional stiffness

Equate $U = \frac{1}{2}K_{\theta,\mathrm{eq}}\theta^2$:

$$\boxed{K_{\theta,\mathrm{eq}} = k_{t1} + k_{t2} + (k_1 + k_2)l_1^2 + k_3 l_2^2.}$$

The restoring torque is $M_{\mathrm{rest}} = -K_{\theta,\mathrm{eq}}\theta$. An applied holding torque of the same magnitude balances it.

4. Check moments and units

A linear spring at radius l_i exerts force $k_i l_i\theta$, giving torque $k_i l_i^2\theta$. This independently explains each squared lever arm. Linear stiffnesses in N/m must be multiplied by squared lengths before they can be added to torsional stiffnesses in N m/rad.